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GROUP SPECIFICATION

Fifth Generation Fixed Network (F5G); Technical Requirements for FTTR Network Deployment

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Fifth Generation Fixed Network (F5G).

Modal verbs terminology

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Introduction

Fiber to the Room (FTTR) technology provides home users with a stable and high-speed network experience by directly laying optical fibers to every room in a building or home, and by efficiently coordinating optical and Wi-Fi® technologies. However, during the large-scale deployment of FTTR, the complexity and the difficulty in troubleshooting network faults often make it challenging to achieve the expected quality standards for FTTR network deployment. As a result, users do not fully enjoy the convenience brought by high-quality FTTR networks in real-world scenarios. They often face issues such as incomplete network coverage, substandard network speeds, severe network interference, high latency, and insufficient operations and maintenance capabilities, which significantly affect user experience. To significantly improve user experience and build truly high-quality FTTR networks, it is particularly necessary to establish standards for the deployment quality of FTTR networks.

1 Scope

The present document specifies the deployment requirements for building high-quality FTTR networks, covering various aspects including FTTR network coverage requirements, network connection requirements, network performance requirements, service experience requirements and network operations and maintenance requirements.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] Recommendation ITU-T G.9940: "High speed fibre-based in-premises transceivers - system architecture".

[2] ETSI TS 103 933 (V1.1.1): "Access, Terminals, Transmission and Multiplexing (ATTM); Fibre deployment for fibre-based in-premises networking".

[3] Recommendation ITU-T G.9941: "High speed fibre-based in-premises transceivers - physical layer specification".

[4] IEEE Std 802.11™-2024: "IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks--Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications".

NOTE: This standard incorporates previous amendments including IEEE Std 802.11ac™-2013, IEEE Std 802.11ax™-2021, IEEE Std 802.11k™-2008, IEEE Std 802.11v™-2011, and IEEE Std 802.11r™-2008. The consolidated standard IEEE 802.11™-2024 is the preferred reference.

[5] Wi-Fi® Alliance: "Wi-Fi EasyMesh® Specification".

[6] IEEE Std 802.1Q™-2022: "IEEE Standard for Local and Metropolitan Area Networks--Bridges and Bridged Networks".

[7] Recommendation ITU-T G.9949: "Wireless local area network management control interface (WMCI) for in-premises networks".

[8] BBF TR-069: "CPE WAN Management Protocol".

[9] IETF RFC 2681: "A Round-trip Delay Metric for IPPM". G. Almes, S. Kalidindi, M. Zekauskas. September 1999.

[10] IETF RFC 3393: "IP Packet Delay Variation Metric for IP Performance Metrics (IPPM)". C. Demichelis, P. Chimento. November 2002.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

installation and maintenance APP: application used by professional installation and maintenance personnel for FTTR network deployment, commissioning, testing and acceptance

primary splitting point: first splitter in the FTTR ODN network located the MFU that splits the optical signal from upper-level network (PON/OLT) to secondary splitter or directly connecting to a limited number (2 - 8 SFUs) of remote terminals.

secondary splitting point: subsequent splitter following after the Primary Splitting Point

NOTE: The secondary splitting point distributes the optical signal to terminals or rooms within a smaller scope.

service serve: remote server providing the service the user is accessing

NOTE: Examples of services include web browsing, gaming, video streaming, cloud applications.

user APP: application used by FTTR subscribers for self-service management of their home network

NOTE: Typical functions include network status viewing, basic Wi-Fi® configuration, one-click fault diagnosis.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

5G	5th Generation, 5GHz Wi-Fi® band
ACK	Acknowledgment
AP	Access Point
APP	Application
ATTM	Access, Terminals, Transmission and Multiplexing
BSS	Basic Service Set
CPU	Central Processing Unit
DFS	Dynamic Frequency Selection
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
DSCP	Differentiated Services Code Point
E2E	End-to-End

FQDN	Fully Qualified Domain Name
FTTH	Fibre-To-The-Home
FTTR	Fibre-To-The-Room
GE	Gigabit Ethernet
GO-GC	Group Owner - Group Client
GPON	Gigabit-capable Passive Optical Network
HD	High Definition
ID	Identifier
IEEE	Institute of Electrical and Electronics Engineers
IFDN	In-premises Fibre Distribution Network
IoT	Internet of Things
IP	Internet Protocol
IPDV	IP Packet Delay Variation
IPTV	Internal Protocol TeleVision
LAN	Local Area Network
MAC	Media Access Control
MFU	Main FTTR Unit
O&M	Operations and Maintenance
ODN	Optical Distribution Network
OLT	Optical Line Termination
P2MP	Point to Multipoint
PAD	Personal Access Device
PC	Personal Computer
PCP	Priority Code Point
PON	Passive Optical Network
QoS	Quality of Service
RSSI	Received Signal Strength Indication
RTT	Round-Trip Time
SFU	Sub FTTR Unit
SME	Small and Medium Enterprise
SN	Serial Number
SNR	Signal-to-Noise Ration
SSID	Service Set IDentification
STA	Station
STB	Set Top Box
TR	Technical Report
VLAN	Virtual Local Area Network
VoIP	Voice over Internet Protocol
VR	Virtual Reality
WAN	Wide Area Network
WLAN	Wireless Local Area Network
WMCI	WLAN Management Control Interface

4 Overview of FTTR Network Deployment Quality

4.1 FTTR Network Deployment Architecture

Based on the G.fin reference model defined in Recommendation ITU-T G.9940 [1], and adapted according to the actual deployment scenario. The FTTR network deployment architecture is achieved through the MFU, SFU, the indoor fibre distribution network, the FTTR management system, the Installation and Maintenance APP, and the User APP. The specific architecture is shown in Figure 1.

NOTE: APP denotes applications installed on off-the-shelve mobile terminals or computers.

FTTR extends fiber optics into every room within the home connected on Fibre-To-The-Home (FTTH), achieving full fiber coverage of the internal home network. The MFU connects upstream to the PON port of the central office OLT equipment and connects downstream to multiple SFU via optical splitters or optical-electrical composite cables. The SFU extend to the user's required deployment areas based on the internal structure, providing wired and wireless coverage for each area, achieving whole-home network coverage. User terminals access the network via Wi-Fi® or Ethernet to meet internet access needs.

The FTTR management system supports one-stop network management services for the FTTR network, enabling visibility, manageability, and maintainability by collecting information such as MFU/SFU operational status, user network configuration information, and network operational status. It supports FTTR network operational status viewing, network quality monitoring, and network fault diagnosis capabilities.

The Installation and Maintenance APP supports FTTR service information display, network configuration parameter distribution, network deployment quality acceptance, and network fault diagnosis capabilities, assisting installation and maintenance personnel in efficient, standardized deployment, testing, and acceptance.

The User APP supports user network status viewing, network function configuration, and network fault diagnosis functions, enabling FTTR users to perform self-service management and maintenance.

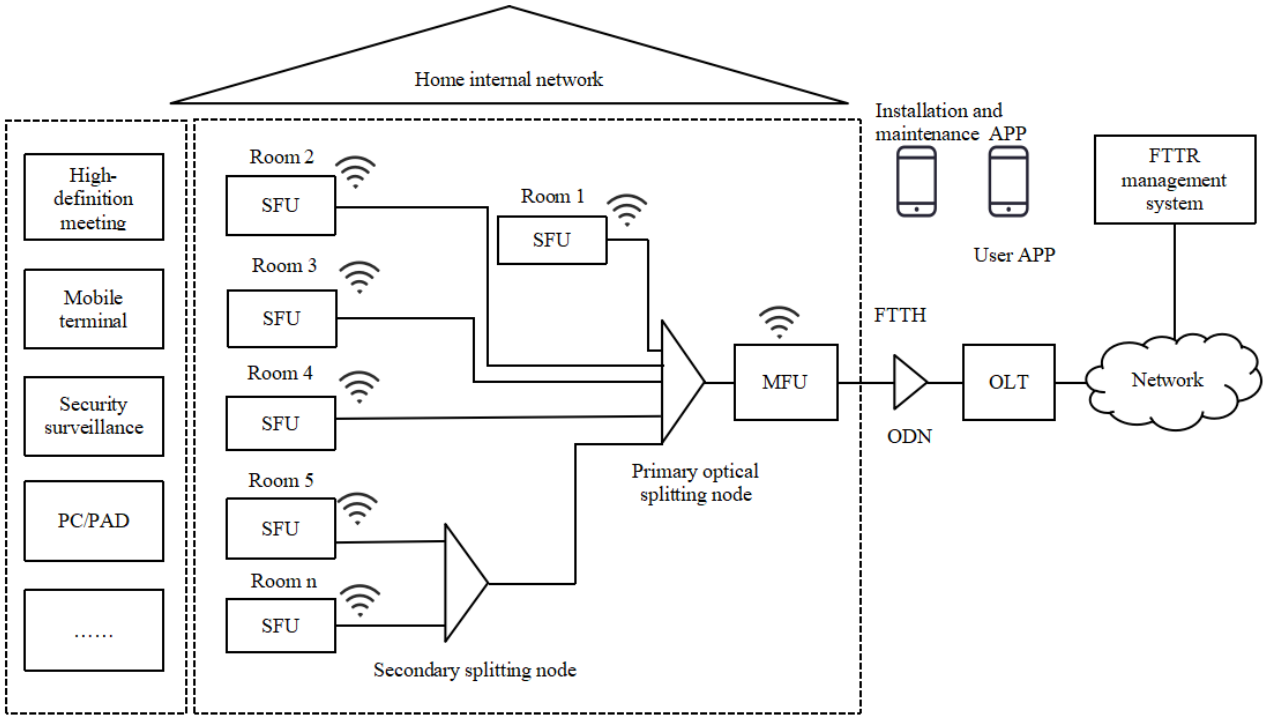


Figure 1: FTTR Network Deployment Architecture

4.2 Architecture of the Evaluation Indicator System

The FTTR networks deployment quality evaluation indicator system consists of five indicator classes: network coverage, network connection, network performance, service experience, and network operations and maintenance. The specific architecture is shown in Figure 2.

a) **Network Coverage:** The network coverage indicators focus on the core dimensions of coverage completeness (such as the proportion of effective coverage throughout the house) and signal quality (such as the Wi-Fi® reception level in each area), aiming for a coverage goal of "no blind spots and strong signals". Ultimately, this ensures that FTTR users enjoy a seamless connection experience of "full-house gigabit" without any interruptions.

b) **Network Connection:** Network connection indicators include network access methods, networking modes, FTTR devices coordination, and network configuration, ensuring the FTTR network has stable basic access capabilities, efficient topology coordination mechanisms, and standardized parameter configuration levels, supporting whole-home quality coverage and reliable service carrying.

c) **Network Performance:** Network performance indicators include network accessibility performance, network rate (the data transfer speed achieved over the network), network latency, network interference, network stability, and seamless roaming, aiming to ensure the FTTR network achieves high-speed, high-reliability connection, low-latency response, low-interference operations, and seamless roaming capabilities in all scenarios.

d) **Service Experience:** Service experience indicators focus on the specific requirements of different types of service for network bandwidth, latency, stability, and roaming, precisely measuring the user's actual perceived service quality. The aim is to translate FTTR's "technical compliance" into the user's "experience compliance", eliminating the cognitive gap where "deployment indicators are qualified but the user feels it is 'not good to use'", ensuring users practically enjoy the real service value of "whole-home gigabit".

e) **Network Operations and Maintenance:** Network operations and maintenance indicators include the functional requirements that the FTTR management system, Installation and Maintenance APP, User APP and the FTTR system shall meet, ensuring the FTTR network achieves efficient deployment processes, stable operational status, and timely fault repair, ultimately supporting high-quality operations and maintenance throughout the network's entire lifecycle, guaranteeing users FTTR network experience.

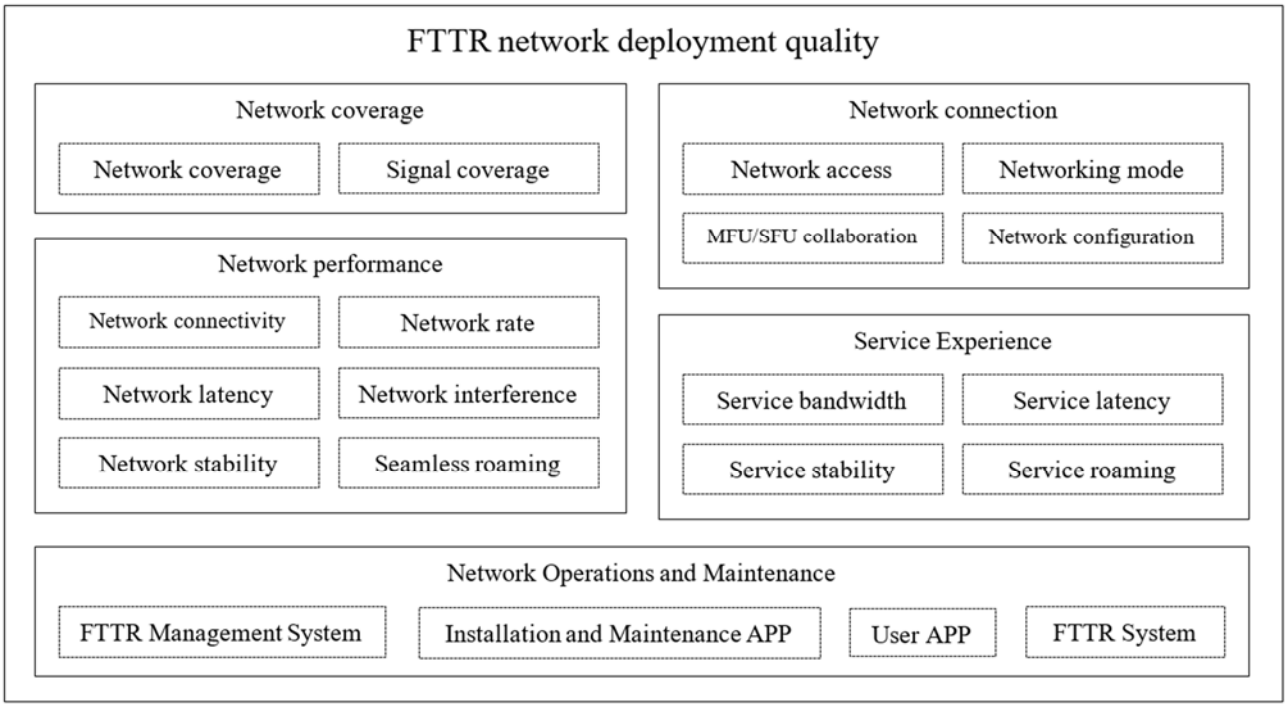


Figure 2: Architecture of the Evaluation Indicator System

5 Network Coverage Requirements

5.1 Coverage Ratio Requirements

To ensure users obtain stable network connections at any location, including areas with elevation differences within multi-story buildings, the building's network coverage shall meet the following requirements:

a) Single-story buildings: The effective coverage area of network signals shall reach 95 % or above of the building's floor area.

b) Multi-story/complex-structure buildings (e.g. duplex residences, commercial buildings, underground garages, etc.):

- Each single floor's area coverage rate shall reach 95 % or above (measured as a 2D area on each floor).

- The overall volume coverage rate of the vertical space shall reach 95 % or above (measured as a 3d volume coverage accounting for stairways, elevators and transitions between floors where the signal may be challenged by multiple structural barriers).

NOTE: A network signal strength of ≥ -70 dBm is the benchmark value for determining qualified coverage.

5.2 Signal Coverage Strength Requirements

Based on the user's actual network usage needs, primary application scenarios, and usage habits, the FTTR network coverage area is divided into core coverage areas and edge coverage areas. The signal coverage strength in different areas shall follow the requirements:

a) Core Coverage Areas: In the main activity areas, where users frequently use the network, such as the living room, master bedroom, and study area, the core coverage scenario signal strength shall be ≥ -65 dBm.

b) Edge Coverage Areas: In areas where users occasionally use the network or have relatively low usage demands (e.g. less than 30 minutes per day of fewer than 3 times per week), such as balconies, bathrooms, and corridors, the edge scenario signal strength shall be ≥ -70 dBm.

When designing the FTTR network deployment plan, the wireless signal attenuation of distance and obstructions are referred to in Annex A and Annex B.

6 Network Connection Requirements

6.1 Network Access Requirements

The home's internal network shall be built as a full optical network system with the MFU as the core. The upstream port of the MFU shall communicate with the public communication network's OLT through the entrance facility. The OLT shall have at least 10G PON network ports to connect the MFU to ensure sufficient bandwidth and high-speed data transmission capabilities. The user indoor network access method is shown in Figure 3.



Figure 3: User Indoor Network Access Method

6.2 Networking Mode Requirements

According to the splitting architecture requirements of Recommendation ITU-T G.9940 [1] and topology of FTTR system requirements of ETSI TS 103 933 [2], FTTR networks shall adopt the Point-to-Multi-Point (P2MP) networking methods. The MFU connects to multiple SFU through one passive splitter (or cascaded extension splitters), with all SFU sharing the backbone fiber bandwidth. Deploying this solution requires meeting strict optical power budget requirements and reasonably planning the split ratio and splitter location. Its architecture is divided into two types:

a) Single Splitter Architecture: One splitter is centrally placed, with a simple structure that is easy to maintain. It is suitable for scenarios with concentrated area and a moderate number of Access Points (SFUs) (typically 4 - 8 SFUs within a single floor layout), as shown in Figure 4.

b) Multiple Splitter Cascaded Architecture: Coverage is extended through optical splitter cascading, which flexibly adapt to complex layouts like duplexes and multi-story homes, but requires higher planning for link loss, as shown in Figure 5.

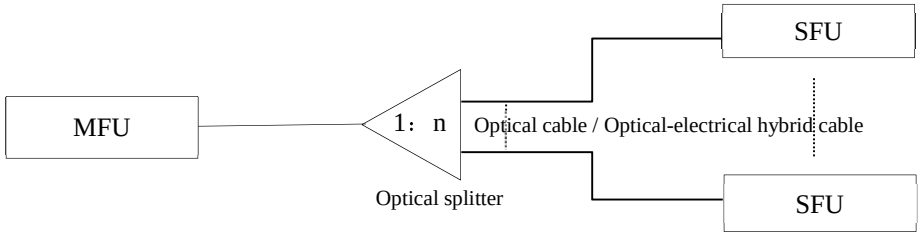


Figure 4: Single Splitter Architecture

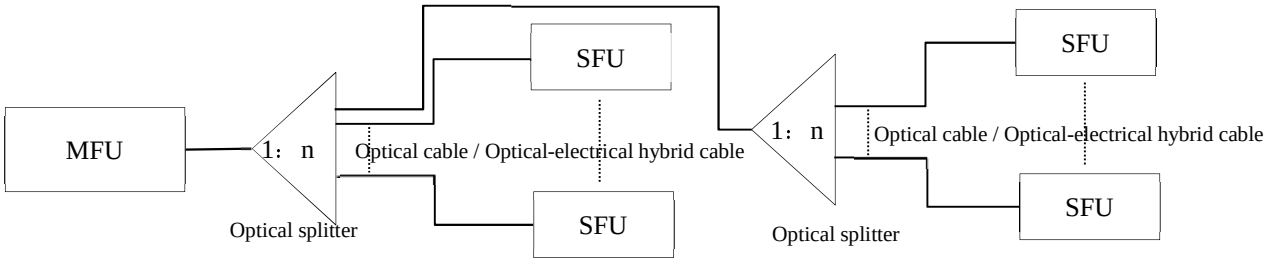


Figure 5: Cascaded Architecture of Multiple Optical Splitters

6.3 MFU/SFU Combination Requirements

6.3.1 Optical Power Budget Requirements

In accordance with the optical interface parameter requirements specified in Recommendation ITU-T G.9941 [3], the MFU and SFUs in an FTTR network shall be selected with matching IFDN classes (e.g. Ra or Rb). This ensures that the received optical power at each SFU stays within the specified sensitivity and overload limits.

To guarantee stable and reliable optical links during deployment, the total loss of the entire optical path shall be accurately estimated based on fiber distance, the number of connectors, and the insertion loss of the splitter in P2MP topologies.

In exceptional cases, like scenarios with FTTR device upgrades, mixed use, or deployment in complex layouts (e.g. multi-level splitting), if the classes are not perfectly matched, engineering methods such as adding fixed optical attenuators, changing the split ratio, or adjusting splitter configurations shall be used for adaptation and calibration to avoid operational risks and ensure long-term network stability and reliability. These cases are operationally complex and therefore are not recommended.

6.3.2 WLAN Protocol Requirements

The deployment environment might involve mixed networking of FTTR devices. This might be the case when the devices are either from different brands, they are from the same brand but different generations, or when the user's existing network contains routers or other network access terminals. To ensure Wi-Fi® signal compatibility and maximize overall network performance, all networking devices within an FTTR network deployment that transmit wireless signals shall support the same Wi-Fi® protocol generation [4] or be downward compatible. This means that the protocol generation chosen for the FTTR network shall be the generation supported by all devices in the network.

6.3.3 Roaming Protocol Requirements

To ensure users obtain a continuous, stable, high-quality wireless network experience during movement, optimize overall wireless network resource utilization, and enhance overall system capacity, in FTTR mixed networking environments, all access point (Aps) shall meet the following roaming protocol requirements:

a) Basic Protocol Level: All APs accessing the FTTR network, including MFU, SFU and pre-existing routers on the user side, shall support and enable the following IEEE 802.11™ [4] protocol enhancements:

   - Radio Resource Measurement of Wireless LANs;

- Wireless Network Management;
- Fast BSS Transition.

b) Networking Coordination Level: On the premise of meeting basic protocol requirements, all APs connected to the FTTR network shall support the same Mesh networking protocol to achieve centralized management and real-time information synchronization between the MFU and SFUs.

EXAMPLE 1: Mesh networking protocols could be the Wi-Fi® EasyMesh™ [5] defined by the Wi-Fi® Alliance, or a proprietary Mesh protocol.

c) Packet Loss Level: To ensure service continuity during roaming handovers, the FTTR network shall support zero-packet-loss roaming.

EXAMPLE 2: Example implementation options include:

- Dual-Link Roaming Mechanism: Utilizing dual-band or dual physical interfaces, this approach establishes connections via two links in advance. During the handover phase, the standby link maintains continuous data transmission.
- Single Physical Interface Roaming Mechanism: This approach establishes a Group Owner - Group Client (GO-GC) communication link between the original AP and the STA. It leverages this link to sustain data forwarding during the interval between the disconnection of the primary link and the establishment of the new link, thereby preventing service interruption.
- Roaming Protocol Enhancement: Optimizing signalling interactions and state synchronization mechanisms to implement a "make-before-break" handover procedure, ensuring lossless transition.
- Centralized Control and Real-Time Data Synchronization: A centralized controller coordinates the STA context and data forwarding among multiple APs, enabling collaborative roaming.

During FTTR network deployment, the protocol support status of all APs shall be audited and confirmed. During APs selection, verification shall be conducted to confirm support for IEEE 802.11™ [4] protocol, Mesh protocol compatibility, and the roaming packet loss mitigation protocol, prioritizing the deployment of a homogeneous system network to minimize interoperability risks and ensure optimal performance.

For existing APs that do not meet the coordination requirements, it is recommended to isolate them by setting up independent networks or adjusting them to bridge/client modes to prevent negative impacts on overall roaming performance and network efficiency.

6.4 Network Configuration Requirements

6.4.1 WAN Configuration Requirements

In FTTR network deployment, the configuration of WAN connections serves as the foundation for ensuring normal service provisioning, stable operation, and efficient management.

The configuration shall adhere to the principles of service isolation, differentiated policies, and security & controllability. In accordance with the service templates or technical specifications provided by the operator, the corresponding WAN connection parameters shall be configured on the FTTR MFU.

Each connection shall implement logical isolation and carry differentiated forwarding policies and security policies.

The following types and requirements of WAN connection configuration are as follows:

a) FTTR Management WAN:

Used for secure communication between the MFU and the operator's FTTR management system, it enables remote automatic deployment, status monitoring, performance management, fault diagnosis and software upgrade of the device.

The FTTR management system server address (IP address or FQDN), authentication information and secure transmission protocols shall be configured. The FTTR devices shall support queries and commands from the specified management domain, while direct access from the Internet side shall be strictly prohibited.

b) Internet Access WAN:

Serves as the core access channel for broadband services, carrying all ordinary Internet data traffic of users.

Mandatory configuration:

- The corresponding encapsulation mode and authentication parameters shall be configured according to the contracted bandwidth and service model.

Optional configuration:

- Multi-WAN Load Balancing: if multiple WAN connections are available - load balancing may be enabled to distribute traffic across connections.
- Link Backup: if redundant WAN connections are available, link backup may be configured to provide failover capability.
- Policy Routing: if traffic needs to be directed based on specific policies, policy routing functions may be configured.
- Traffic Shaping and Connection Limits: traffic shaping, connection limits and default security policies may be configured as needed to manage network resources.

c) Video Service WAN:

Used to carry the operator's IPTV video traffic. Multicast protocol parameters and high-priority QoS policies shall be configured to ensure effective distribution of video streams within the home network and efficient multicast forwarding in the FTTR system.

d) Voice Service WAN:

Used to carry VoIP traffic. A dedicated VoIP access connection shall be established, with the corresponding voice VLAN, server address and authentication information configured. The highest-priority Differentiated Services Code Point (DSCP) marking shall be enabled, and top-priority queue scheduling and forwarding assurance shall be guaranteed across the entire network to minimize delay and jitter.

6.4.2 VLAN Configuration Requirements

During FTTR network deployment, VLAN division and priority configuration shall be implemented according to service type, user type, physical ports and other factors. By isolating traffic of different services, network security shall be enhanced and network management efficiency shall be optimized. Specific requirements are as follows:

a) Division by service: Independent service VLANs shall be assigned to different upstream services to implement Layer 2 logical isolation of traffic.

b) On-demand intranet isolation: Different user VLANs shall be divided on the user side (LAN/Wi-Fi®) as required.

EXAMPLE: For example, smart home devices and guest terminals are isolated from the main home network to improve security.

c) Binding to ports and SSIDs: Each VLAN shall be bound to the corresponding physical port, Wi-Fi® SSID or specific SFU, ensuring that service traffic is accurately directed to target terminals.

d) QoS priority marking: In the 802.1Q [6] tag of a VLAN, the Priority Code Point (PCP) value shall be configured according to service sensitivity, providing a foundation for end-to-end quality of service assurance.

e) Planning and synchronization: VLAN IDs shall be uniformly planned before deployment to avoid conflicts. During configuration, VLAN information shall be synchronized across the entire network between the MFU and SFU devices to ensure service continuity.

6.4.3 WLAN Configuration Requirements

6.4.3.1 Channel Allocation Requirements

In an FTTR network, reasonable channel allocation and transmit power control shall be adopted to maximize system capacity and user experience, so as to achieve minimized interference, load balancing and dynamic optimization of the FTTR network. The specific requirements are as follows:

Requirements for Band-Specific Differentiated Configuration

- 2,4 GHz Band: Owing to the scarcity of available non-overlapping channels (usually only channels 1, 6 and 11) and high vulnerability to interference from non-Wi-Fi® devices such as Bluetooth® and microwave ovens, the 2,4 GHz band features a crowded channel environment.

  To realize effective interference avoidance and stable roaming, all SFUs shall operate on the same channel in the 2,4 GHz band, so as to minimize mutual interference with neighbouring APs in dense residential scenarios and convert competition among multiple internal APs into controllable cooperative scheduling.

  Given its technical characteristics, this band is applicable for the following scenarios:

  - IoT and smart home connections
  - Legacy or 2,4 GHz-only terminal devices
  - Services with low bandwidth and high delay tolerance
  - Supplementary coverage for edge areas where 5 GHz signals do not achieve stable coverage
  - Residential areas with highly dense Wi-Fi® environments

- 5 GHz / 6 GHz Bands: The 5 GHz and 6 GHz bands, characterized by abundant channel resources and relatively low interference, are the key bands in FTTR networks for carrying high-speed services and guaranteeing overall FTTR system capacity.

  To achieve efficient collaboration and optimal performance among multiple access points, the following requirements shall be followed during deployment:

  - Channel Isolation: Different non-overlapping 5 GHz / 6 GHz channels shall be assigned to adjacent SFUs or those with overlapping signal coverage, to realize spatial reuse, reduce co-channel interference and improve overall FTTR system throughput.
  - High-Performance Channel Preference: Channels with low interference shall be prioritized. For the 5 GHz band, channels supporting Dynamic Frequency Selection (DFS) (e.g. channels 52 - 144) shall be considered to utilize wider continuous spectrum, on the premise that devices comply with relevant radar detection specifications.
  - Load and Interference Awareness: Devices shall support a dynamic channel switching mechanism based on real-time load and ambient noise; alternatively, static optimized planning shall be performed via site survey tools in the initial deployment stage.

Requirements for Channel Width Configuration

- 2,4 GHz Band: a channel width of 20 MHz should ensure extensive signal coverage, satisfactory WLAN terminals compatibility and connection stability.
- 5 GHz / 6 GHz channels shall be assigned to adjacent SFUs or those with overlapping signal coverage, to realize spat 6 GHz Bands: Channel widths of 20 MHz, 40 MHz, 80 MHz or 160 MHz are flexibly selected according to WLAN terminal capabilities, environmental interference levels and service requirements.
- 80 MHz: As the mainstream configuration shall be available, as it strikes a good balance between ultra-high throughput (suitable for 4K streaming and large file transmission) and certain anti-interference capability. It shall be ensured that adjacent APs use different 80 MHz channel blocks.

- 160 MHz: It delivers ultimate rates and is an ideal choice for Wi-Fi® 6E/7 devices in the 6 GHz band or open 5 GHz environments. However, it occupies an extremely wide spectrum and imposes stringent requirements on link quality, making it highly prone to conflicts with surrounding networks in densely deployed scenarios.

NOTE: It is only for cautious use in independent environments with minimal interference or specific high-speed application scenarios.

- Dynamic Adjustment: advanced FTTR systems shall support a dynamic bandwidth adjustment mechanism based on channel interference conditions. When strong interference is detected in adjacent channels, the FTTR system automatically switches from 80 MHz to 40 MHz to improve connection stability.

Automation Requirements

The MFU and SFU shall implement dynamic channel selection and optimal allocation based on the WMCI protocol [7]. The MFU shall automatically or periodically assign optimal channels and transmit power to all SFUs in a unified manner, and adjust dynamically according to environmental changes, so as to achieve global optimization and efficient management of wireless resources.

The FTTR system shall support the manual channel locking function to meet the planning requirements of special scenarios.

EXAMPLE: In cases of fixed channels for wireless backhaul.

6.4.3.2 Roaming Configuration Requirements

The configuration of the roaming threshold (generally referring to the minimum Received Signal Strength Indicator threshold that triggers terminal handover) shall be planned according to the physical structure, wall material and signal attenuation characteristics of the actual deployment environment, so as to achieve the optimal balance among coverage continuity, network stability and service performance.

As an initial reference for engineering implementation, differentiated baseline thresholds shall be set for different frequency bands:

- 2,4 GHz band: Due to its low penetration loss and wide coverage range, the baseline RSSI threshold for roaming triggering shall be -65 dBm.

- 5 GHz / 6 GHz band: Due to its high penetration loss and clear coverage boundary, the baseline RSSI threshold for roaming triggering shall be -60 dBm to facilitate more active handover.

To avoid ping-pong roaming caused by an excessively high threshold and terminal stickiness caused by an excessively low threshold, professional tools shall be used for on-site roaming testing during FTTR network deployment. The threshold shall be finely tuned and optimized based on test results including roaming handover points, packet loss rate and service experience, to determine the optimal value for the specific environment.

In multi-story, duplex or complex-structure deployment scenarios, differentiated threshold policies shall be adopted for different areas or floors, or FTTR network functions supporting dynamic threshold adjustment shall be enabled.

6.4.4 DHCP Configuration Requirements

It serves as the foundation for automatic terminal access, orderly network management and correct service bearing. The DHCP service configuration shall comply with the following requirements:

a) **DHCP Service Mode**

MFU: The DHCP service shall be enabled to build an IPv4/IPv6 dual-stack network. As the management and control core of the user's intranet, it shall implement unified centralized addressing and network parameter configuration for all terminals.

SFU: The local DHCP service shall be forcibly disabled. The SFU shall operate in bridge mode or controlled access point mode, and be only responsible for transparently forwarding DHCP protocol packets delivered by the MFU, so as to avoid multiple DHCP server conflicts in the network.

b) **IP Address Pool Allocation**

Core principle: The address pool size shall fully consider the current number of terminals and reserve sufficient space for future expansion. Meanwhile, the address segment shall not conflict with WAN-side service addresses or networking devices management addresses.

IPv4 addresses: An appropriate private address segment shall be planned, with the scope and excluded addresses of the DHCP address pool clearly defined.

IPv6 addresses: An IPv6 global unicast prefix shall be obtained from the upstream and correctly advertised. The MFU shall be able to proxy this prefix to ensure that all SFUs and terminals generate valid IPv6 addresses.

Static addresses: For key devices requiring fixed IP addresses, static address binding shall be configured according to their MAC addresses to ensure constant IP addresses for easy access and management.

c) **Other Parameter Configuration**

Gateway address: The delivered default gateway address shall be set to the LAN interface IP address of the MFU.

DNS servers: The DNS server addresses specified by the operator shall be preferentially configured. Well-known public DNS servers may also be added as backups to ensure the efficiency and reliability of domain name resolution.

Address lease: Differentiated strategies shall be formulated based on specific network scenarios to balance address utilization and network stability.

- For typical home environments, the lease time shall be set to 24-72 hours to reduce frequent lease renewal requests from terminals, thereby lowering FTTR devices energy consumption and network overhead while improving roaming stability.
- For fixed terminals in home networks, static IP binding shall be adopted to prevent service interruptions caused by IP address changes.
- For high-mobility scenarios, the lease time shall be set to 1-2 hours to enable rapid recycling and reuse of the IP address pool, preventing new users from failing to connect due to address exhaustion.

7 Network Performance Requirements

7.1 Network Connectivity Requirements

7.1.1 Physical Layer Connectivity Requirements

Optical power management serves as the physical layer foundation for ensuring the stable and reliable operations of optical fiber links and the normal functioning of equipment. In an FTTR system, the loss and intensity of optical signals during transmission shall be controlled to keep the optical power at the transmitting and receiving ends strictly within the normal operating range of the equipment. This prevents the saturation or even damage of receiving devices caused by excessively high optical power, as well as connection interruptions due to overly low optical power. The specific requirements are as follows:

a) Transmit optical power requirement: The transmit optical power of MFU/SFU shall not exceed the maximum transmit optical power limit supported by the MFU/SFU.

b) Receive optical power requirement: The receive optical power of MFU/SFU shall not be lower than the nominal receive sensitivity of the MFU/SFU.

c) Link loss requirement: The total FTTR link loss (including fiber transmission loss, splitter loss, and connector/splice point loss) shall not exceed the difference between the MFU transmit power and the SFU receive sensitivity.

NOTE: This difference represents the maximum allowable attenuation threshold for the link.

7.1.2 Service Connectivity Requirements

Service reachability requirements mean that all types of terminals connected to the FTTR network shall stably and reliably access their corresponding upper-layer service platforms and applications. The following service types and their requirements of service connectivity are as follows:

a) Internet access service: User terminals connected to the FTTR network shall successfully resolve domain names and access Internet resources, or directly establish connections with external networks via public network IP addresses.

b) IPTV service: After a STB/TV set accesses its designated exclusive port or wireless network, it shall automatically obtain an IP address dedicated to IPTV services, and stably play live channels and video-on-demand content.

c) VoIP service: A voice gateway or IP phone shall successfully register with the operator's VoIP platform, and stably make and receive calls to ensure the quality of voice communication.

7.1.3 Management Channel Connectivity Requirements

The reachability of the management channel is the foundation for ensuring the remote O&M and long-term stable operations of the FTTR network. The FTTR network shall support the management of the BBF TR-069 [8] protocol at a minimum, with the specific requirements as follows:

a) BBF TR-069 [8] Management: The MFU shall be able to establish and maintain a stable connection with the operator's FTTR management system. After the MFU is successfully registered on the FTTR management system, it shall automatically complete the periodic reporting of FTTR devices information, and reliably receive and execute the configuration commands and software upgrade tasks issued by the FTTR management system, so as to realize the remote centralized management of the network.

7.2 Network Rate Requirements

7.2.1 PON Access Rate Requirements

To ensure that the FTTR system does stably carry the broadband services subscribed by users and provide a high-quality network experience for the entire house, the actual available receive rate of the uplink PON ports of MFU and SFU shall be no less than 100 % of the nominal rate of the broadband package subscribed by users.

To address protocol overhead, instantaneous traffic bursts and long-term performance fluctuations, the PON port link rate capability of the MFU shall have a design margin of 20 % or more higher than the subscribed rate, so as to ensure that the subscribed rate is stably achieved under various load conditions.

7.2.2 Wi-Fi® Rate Requirements

7.2.2.1 Single Terminal Rate in Core Area

a) Terminal near-field rate: Within a 1-meter range of the MFU and SFU access points in an unobstructed environment, the 5 GHz band shall be prioritized for rate testing, and the measured maximum download rate shall be no less than 90 % of the nominal rate of the user's subscribed broadband package.

b) Simulation of real user scenarios: Rate testing shall be conducted in typical WLAN usage areas such as the living room, bedroom and study in a normal usage posture (e.g. sitting or standing), with no artificial abnormal obstruction to the signal path between the terminal and the MFU/SFU. The 5 GHz band shall be prioritized for the test, and the measured maximum download rate shall be no less than 85 % of the nominal rate of the user's subscribed broadband package.

7.2.2.2 Single Terminal Rate in Edge Area

In edge coverage areas such as balconies, kitchens and bathrooms with 1 to 2 ordinary wall obstructions from the MFU or SFU, the 5 GHz band shall be prioritized for rate testing, and the measured maximum download rate shall be no less than 50 Mbps to meet the usage requirements of basic services.

7.2.2.3 Multi-terminal Concurrent Rate

In the scenario of multi-terminal concurrent access to the FTTR network, the bandwidth allocation for each terminal shall be guaranteed to be fair, stable and free from obvious degradation. When all terminals are online simultaneously and running typical data services (including but not limited to web browsing, video playback, file downloading, voice/video calls, online gaming and smart home control, etc.), the service experience of each terminal shall have no phenomena such as freezing, disconnection, sharp latency spikes or abrupt rate drops.

7.3 Network Stability Requirements

7.3.1 Network Latency Requirements

7.3.1.1 In-Premises Network Latency Requirements

The in-premises segment is defined as the path from the user terminal to the MFU uplink port (i.e. the boundary between the in-premises FTTR network and the external operator network). These metrics are used for fault isolation and verification of in-premises network quality, independent of external network conditions. The Round-Trip Time (RTT) [9] within this segment shall meet the following requirements:

a) Within the core coverage area, the average RTT shall be $\leq$ 20 ms;

b) Within the edge coverage area, the average RTT shall be $\leq$ 35 ms.

7.3.1.2 End-to-End Network Latency Requirements

The RTT is measured from the user terminal, through the FTTR network (MFU/SFU), to the service server and back. The resulting RTT shall meet the following requirements:

a) Within the core coverage area, the average RTT shall be $\leq$ 50 ms;

b) Within the edge coverage area, the average RTT shall be $\leq$ 100 ms.

7.3.2 Network Jitter Requirements

7.3.2.1 In-Premises Network Jitter Requirements

The scope of this clause is the same as that of clause 7.3.1.1. Jitter, defined as IP Packet Delay Variation (IPDV) per IETF RFC 3393 [10] quantifies the variation in packet transit delay across the in-premises network segment. The RTT within this segment shall meet the following requirements:

a) Within the core coverage area, the average RTT shall be $\leq$ 10 ms;

b) Within the edge coverage area, the average RTT shall be $\leq$ 15 ms.

7.3.2.2 End-to-End Network Jitter Requirements

The scope of this clause is the same as that of clause 7.3.1.2. The resulting RTT shall meet the following requirements:

a) Within the core coverage area, the network jitter shall be $\leq$ 20 ms;

b) Within the edge coverage area, the network jitter shall be $\leq$ 30 ms.

7.3.3 Network Packet Loss Rate Requirements

7.3.3.1 In-Premises Network packet loss rate Requirements

The scope of this clause is the same as that of clause 7.3.1.1. The network packet loss rate within this segment shall meet the following requirements:

a) Within the core coverage area, the network packet loss rate shall be ≤ 1,0E-6;

b) Within the edge coverage area, the network packet loss rate shall be ≤ 1,0E-5.

7.3.3.2 End-to-End Network packet loss rate Requirements

The scope of this clause is the same as that of clause 7.3.1.2. The network packet loss rate shall meet the following requirements:

a) Within the core coverage area, the network packet loss rate shall be ≤ 1,0E-5;

b) Within the edge coverage area, the network packet loss rate shall be ≤ 1,0E-3.

7.3.4 Wi-Fi® Coverage Continuity Requirements

When a user terminal moves indoors, the fluctuation range of the Wi-Fi® signal strength to which it is connected shall be no more than 15 dBm, so as to ensure the continuity of Wi-Fi® coverage and the stability of user experience.

7.4 Network Interference Requirements

7.4.1 AP Interference Requirements

To prevent excessive signal interference, overlapping coverage areas and severe over-the-air collisions caused by MFUs, SFUs and other APs being placed too close to each other in physical space, which would degrade connection stability, roaming performance and overall user network experience, the following signal strength isolation requirements shall be satisfied between MFUs and SFUs in an FTTR system:

a) In the FTTR system, the wireless signal RSSI received by an MFU from any SFU or other AP within the system shall be no higher than -60 dBm.

b) In the FTTR system, the wireless signal RSSI received by an SFU from any other SFU, MFU or other AP within the system shall be no higher than -60 dBm.

7.4.2 Wi-Fi® Interference Requirements

In FTTR network deployment, Signal-to-Noise Ratio (SNR), as a core indicator for measuring wireless signal quality, directly determines network transmission performance and user experience. To ensure network performance in different coverage areas, the following requirements are specified for SNR:

a) In the core coverage area, the SNR shall be ≥ 25 dB to meet the requirements of high-density services (such as 4K video streaming, online collaboration, and smart home linkage) for stable and high-speed network signal transmission, and to avoid rate fluctuations and experience degradation;

b) In the edge coverage area, the SNR shall be ≥ 18 dB to ensure continuous signal connection, prevent service freezing and disconnection, meet basic network usage requirements such as web browsing, instant messaging, and lightweight IoT control, and maintain the minimum level of network availability.

7.4.3 Electromagnetic Interference Requirements

When common interference sources such as induction cookers, microwave ovens, Bluetooth® devices and wireless cameras are operating, wireless signals shall not show obvious degradation, frequent disconnections, abrupt rate drops, service interruptions and other abnormal phenomena to ensure stable network operations.

In an electromagnetic interference environment, if the SNR ≤ 15 dB, Wi-Fi® channel assignment or AP location shall be adjusted accordingly.

7.5 Roaming Requirements

To ensure service continuity for users moving between multiple rooms in a home scenario, the FTTR system shall meet the following technical requirements for seamless roaming:

a) Within the core coverage area, the E2E handover latency shall be ≤ 50 ms, the E2E packet loss rate shall be less than 1,0E-5;

b) Within the edge coverage area, the E2E handover latency shall be ≤ 100 ms, the E2E packet loss rate shall be less than 1,0E-3.

8 Service Experience Requirements

To ensure actual user service experience, the following services shall meet the specified indicator requirements during FTTR network deployment.

Table 1: Home Service Experience Deployment Requirements

Service Type	Bandwidth Requirements	E2E Latency Requirements	E2E Jitter Requirements	E2E Packet Loss Rate Requirements	Roaming Requirements
——Voice ——Web Browsing ——Upload /Download ——SD & HD Video	≥ 100 Mbps	≤ 100 ms	≤ 100 ms	≤ 1,0E-3	≤ 100 ms
——4K Video ——Mobile Gaming ——Online Education/ Remote Work	≥ 1 000 Mbps	≤ 50 ms	≤ 50 ms	≤ 1,0E-5	≤ 50 ms
——8K Video ——Cloud Gaming ——Cloud VR	≥ 2 000 Mbps	≤ 20 ms	≤ 15 ms	≤ 1,0E-6	≤ 10 ms

9 Network Operations and Maintenance Requirements

9.1 FTTR Management System Requirements

9.1.1 Network Information Visualization Requirements

The FTTR management system shall support visualization capabilities for the operational status of MFU/SFU and the FTTR network state, in order to realize real-time monitoring and assessment analysis of network quality. The system shall support the collection and display of the following information:

a) **MFU/SFU information:**

- MFU/SFU model
- MFU/SFU MAC
- MFU/SFU online status
- MFU/SFU runtime duration
- MFU/SFU CPU utilization

- MFU/SFU memory utilization
- MFU/SFU optical module temperature
- MFU/SFU transmit optical power
- MFU/SFU received optical power
- MFU/SFU supply voltage
- MFU/SFU bias current
- MFU/SFU upstream operation mode
- Combo MFU operating mode

b) **FTTR network status information:**

- MFU upstream OLT PON port type
- FTTR network topology
- SSID name and password
- Wi-Fi® RSSI
- Wi-Fi® frequency
- Wi-Fi® bandwidth
- Wi-Fi® upload/download rates
- Wi-Fi® roaming strategy (on/off status, handover threshold)
- LAN port rate (negotiated rate, actual rate)
- Cumulative upstream/downstream traffic

9.1.2 Network Parameter Configuration Requirements

The FTTR management system shall support remote and local network parameter configuration for MFU/SFU, including the following configurations:

- WAN configuration
- VLAN configuration
- WLAN configuration
- LAN configuration
- QoS configuration
- DHCP configuration
- Security strategy configuration

9.1.3 Network Fault Diagnosis Requirements

The FTTR management system shall support network fault diagnosis capabilities. The system shall analyse user subscription information, MFU/SFU operational status, and network performance indicators to identify and analyse FTTR network faults. It shall support the following fault identification capabilities:

a) **MFU/SFU operational faults:**

- MFU/SFU offline

- MFU/SFU optical signal unregistered
- MFU/SFU optical module temperature too high
- MFU/SFU received optical power too low or transmitted optical power too high
- MFU/SFU power supply voltage too low or too high
- MFU/SFU bias current insufficient or too high
- MFU/SFU CPU utilization too high
- MFU/SFU memory utilization too high
- MFU upstream OLT lacks 10G PON resources
- Combo MFU failed to switch to 10G PON operation mode

b) **Line connection faults:**

- Loose contact, bending, or breakage of MFU upstream fiber optic cable
- Loose contact, bending, or breakage of SFU upstream fiber optic cable/ethernet cable
- MFU and SFU connected via 100 Mbps ethernet cable, resulting in LAN port negotiation rate below 1 Gbps
- SFU connected to MFU via Wi-Fi®, leading to poor network signal transmission quality

c) **Network configuration faults:**

- WAN configuration error
- VLAN configuration error
- WLAN configuration error
- DHCP configuration error
- QoS configuration requires optimization
- Security strategy configuration requires optimization

d) **Poor network performance:**

- MFU/SFU PON port rate lower than subscribed package rate
- Wi-Fi® real-time download/upload rate too low
- Wi-Fi® 5G band not enabled
- Wi-Fi® channel congestion with severe interference
- Wi-Fi® latency and jitter too high
- Excessive number of terminals connected under MFU/SFU
- Wi-Fi® roaming function not enabled
- Wi-Fi® roaming threshold set too low or too high

e) **Poor service experience:**

- IPTV service unavailable
- IPTV service buffering or loading
- VoIP service connection failure

- IoT service connection failure
- Poor experience of real-time services
- Poor experience of cloud applications

9.2 Installation and Maintenance APP Requirements

The Installation and Maintenance APP is used to assist maintenance personnel in efficient, standardized deployment, commissioning, and acceptance. It shall support the following functions:

a) **FTTR service information inquiry:**

- FTTR user information (username, contact information, installation address, etc.)
- FTTR package information (package name, subscribed rate, package activation time, etc.)
- MFU/SFU issuance information (MFU/SFU MAC, SN, model)
- FTTR installation & maintenance personnel information (personnel ID, contact information, etc.)
- FTTR service acceptance information (service acceptance time, scheduled on-site installation time, service completion time)

b) **FTTR network deployment configuration:**

- MFU/SFU binding and registration
- Network parameter configuration (WAN, VLAN, WLAN configuration, etc.)
- Service parameter configuration (IPTV/VoIP service configuration, etc.)

c) **FTTR network deployment quality acceptance:**

- Network coverage rate (planar coverage rate, spatial coverage rate)
- MFU/SFU optical power (transmit optical power, receive optical power)
- Network connectivity (network access, service access, management channel establishment)
- Network rate (PON access rate, Wi-Fi® rate)
- Network performance (latency, jitter, packet loss rate, interference, roaming)
- Service experience (bandwidth, latency, jitter, packet loss rate, roaming)

d) **On-site fault diagnosis:**

- MFU/SFU working status diagnosis
- Line status diagnosis
- Network configuration diagnosis
- Network performance diagnosis
- Service experience diagnosis
- Fault handling solution recommendation

9.3 User App Requirements

The User APP is for FTTR subscriber self-service management and maintenance. It shall support the following functions:

a) **FTTR network status inquiry:**

- MFU/SFU online status
- MFU/SFU runtime duration
- FTTR network topology
- Number of terminals connected under MFU/SFU
- Cumulative upstream/downstream traffic
- Real-time upstream/downstream rate

b) **FTTR network management:**

- Wi-Fi® switch setting
- SSID name setting
- SSID password setting
- Wi-Fi® band setting
- Wi-Fi® bandwidth setting
- Wi-Fi® roaming setting
- Guest network setting
- Parental control/terminal rate limiting

c) **FTTR network quality report:**

- MFU/SFU PON port rate test results
- Wi-Fi® rate (average upload/download rate, maximum upload/download rate)
- Wi-Fi® latency (average latency, maximum latency)
- Wi-Fi® jitter (average jitter, maximum jitter)
- Wi-Fi® packet loss rate (average packet loss rate, maximum packet loss rate)
- Network interference analysis (AP interference, Wi-Fi® interference, electromagnetic interference)
- Connected terminal type and online duration (mobile, PAD, PC, etc.)
- Service quality analysis (IPTV service, VoIP service, IoT service, real-time service, cloud application, etc.)
- Network quality score

d) **One-click fault diagnosis:**

- Account arrears and status check
- MFU/SFU working status diagnosis
- Line status diagnosis
- Network configuration diagnosis

- Network performance diagnosis
- Service experience diagnosis
- Self-service fault troubleshooting guide
- One-click fault repair request function

9.4 FTTR Protection Requirements

9.4.1 Dual-MFU Protection

The FTTR system consists of MFU and SFU devices. For specific deployment scenarios requiring high availability and service protection, such as industrial networks, the MFU in FTTR protection scheme should include two MFU devices as mutual backup. As shown in Figure 6, the MFU -1 is Active state and MFU-2 is Standby state. The MFU-1 and MFU-2 connect to the SFU devices through full optical fibre with a 2:N optical splitter respectively. An air interface communication link for message exchange should be set up between MFU-1 and MFU-2. Note that the air interface communication link should adopt an extension of IEEE 802.11, the specific implementation is out of scope of the present document.

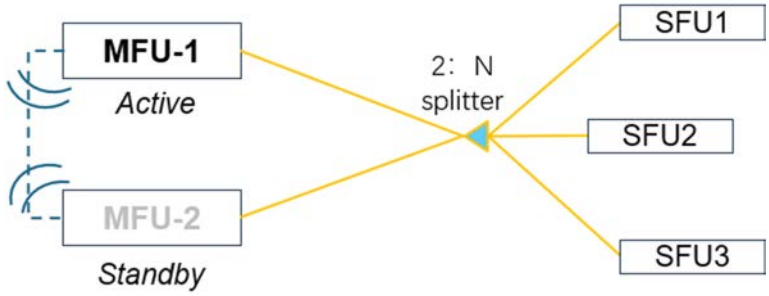


Figure 6: Example of Type-B protection in FTTR system

The protection configuration allows to mitigate a fault in the main fibre (i.e. the fibre segment between the MFU and the splitter) as well as a failure of the MFU interface itself. In case of a main fibre fault or an MFU fault, the handover of all corresponding SFUs should be processed to maintain the normal operation of the system.

For MFU-1, when it operates in the Active state, it should synchronize the information of the FTTR system status (including networking configuration information and SFU status information) to the MFU-2 through the air interface communication link, as follow:

- The MFU-1 should collect the FTTR networking configuration information of the FTTR system in accordance with a preset first synchronization period, generate a configuration synchronization message including the collected FTTR networking configuration information and transmit the configuration synchronization message to the MFU-2 through the air interface communication link. The MFU-2 shall cache the FTTR networking configuration information carried in the received configuration synchronization message to its local storage.
- Similarly, the MFU-1 should collect the status information of all registered SFUs in the FTTR system, to obtain the SFU status information of the system in accordance with a preset second synchronization period, generate a status synchronization message including the collected SFU status information, transmit the generated status synchronization message to the MFU-2 through the air interface communication link. The MFU-2 shall cache the SFU status information carried in the received status synchronization message to its local storage.

The illustration of the FTTR protection switching process is shown in Figure 7. Upon detection of a fault in the optical fibre link between the MFU-1 and the optical splitter by MFU-1, the MFU-1 shall transmit a switching request message to the MFU-2 through the air interface communication link. The switching request message is to instruct the MFU-2 to execute protection switching process based on the system status information cached in its local storage. The MFU-2 shall run the protection switching process upon receipt of the switching request message, which is to switch itself from the Standby state to the Active state. The MFU-1 should receive a switching ACK message from MFU-2 through air interface communication link, which should be sent once the available state of fiber optical link between MFU-2 and the SFUs has been verified. Upon receipt of the switching ACK message, the MFU-1 shall respond by switching itself from the Active state to the Inactive (Standby) state.

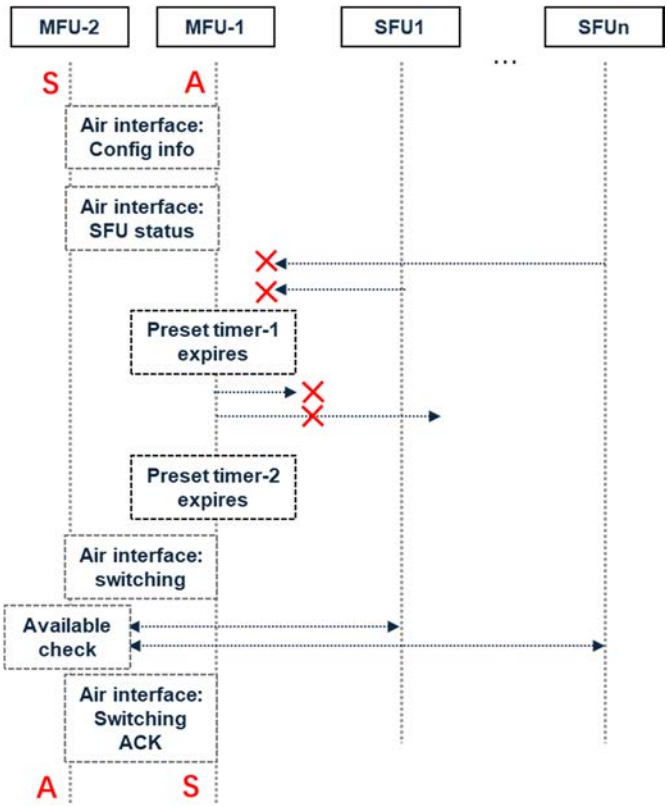


Figure 7: FTTR protection switching process

If no upstream signal is received by the MFU-1 within a preset timer-1, the MFU-1 shall broadcast a keepalive query message to all registered SFUs in the FTTR system. The keepalive query message shall instruct the receiving SFUs to return a keepalive response message to the MFU-1 within the preset timeout period. If none of the registered SFUs returns the corresponding keepalive response message to the MFU-1 within a preset timer-2, the MFU-1 shall confirm that a fault has occurred in the fiber-optic link between itself and the optical splitter. Note that the practical limit of each preset timer for failure detection could be different based on specific application scenarios.

Moreover, the message exchange link between MFU-1 and MFU-2 is not limited to the air interface. For deployment scenarios where wired connections are available or preferred, Ethernet cables or optical fibers are used as communication links:

- Ethernet cable link, as shown in Figure 8: The two MFU devices should be directly connected via their integrated Ethernet electrical ports (typically GE/10GE ports). Message transmission between MFUs should be based on standard Ethernet protocols. This method provides stable, low-latency communication with strong immunity to electromagnetic interference, and is suitable for scenarios where the two MFUs are deployed in close proximity (e.g. same cabinet, same equipment room).

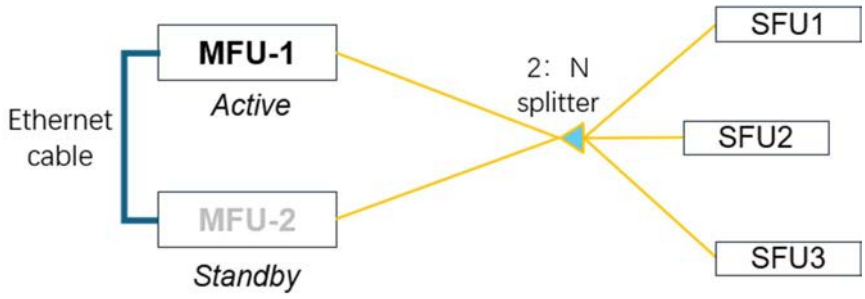


Figure 8 Type-B protection in FTTR system (Ethernet cable based)

- Optical fiber link, as shown in Figure 9: The two MFU devices should be connected via their extra optical ports using single-mode optical fiber. Message transmission should be based on PON protocols. This method offers more flexible cabling and electromagnetic interference immunity, and is suitable for SME (small and medium enterprise) scenarios that have strict requirements on interior appearance while demanding fast redundant protection.

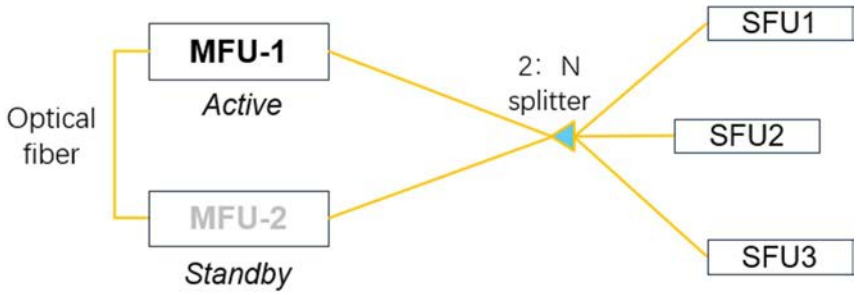


Figure 9: Type-B protection in FTTR system (Optical fiber based)

9.4.2 Single MFU Port-level Protection

In addition to the dual-MFU mutual backup protection architecture, a single-MFU port-level protection architecture is also supported, as shown in Figure 10. In this architecture, a single MFU device is equipped with two independent FTTR ports, which are respectively connected to the two input ends of the 2:N optical splitter through separate optical fiber links. One port operates in the Active state as the primary port, and the other operates in the Standby state as the backup port.

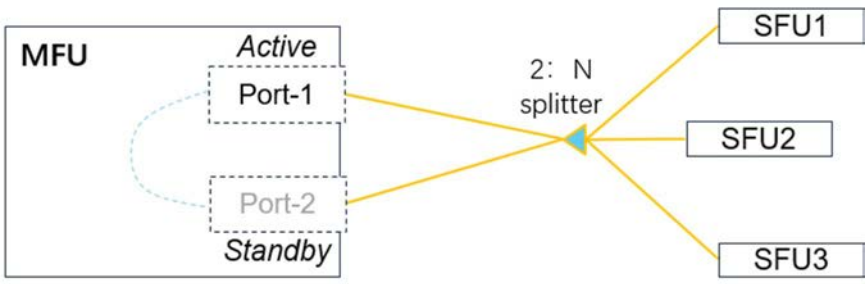


Figure 10: Type-B protection in FTTR system (MFU port-level)

The MFU shall continuously monitor the optical signal status and link connectivity of the primary port. Upon detection of a fault in the primary port or its connected optical fiber link, the MFU shall perform local port switching directly without cross-device message exchange, and switch all service traffic to the backup port. Since all system configuration information and SFU status information are stored locally in the same MFU, no inter-device data synchronization is required, resulting in lower protection switching latency.

Annex A (informative): Reference for Wireless Signal Attenuation by Distance

Table A.1 lists the wireless signal attenuation for different distances without any obstacles.

Table A.1: Reference for Wireless Signal Attenuation by Distance

Distance (m)	2,4 GHz signal attenuation (dB)	5 GHz signal attenuation (dB)
1	40,04 - 40,45	46,20 - 47,78
2	46,05 - 46,46	52,21 - 53,79
3	49,58 - 49,99	55,74 - 57,32
5	54,02 - 54,43	60,18 - 61,76
7	56,94 - 57,35	63,10 - 64,68
10	60,04 - 60,45	66,20 - 67,78
15	63,56 - 63,97	69,72 - 71,30
20	66,05 - 66,47	72,21 - 73,79

NOTE: The above values are theoretical calculations. Deviations may occur in practical applications; actual test results are recommended to be used for verification.

Annex B (informative): Reference for Wireless Signal Attenuation by Obstacles

Table B.1 lists wireless signal attenuation with different material and different thickness levels of the material.

Table B.1: Reference for Wireless Signal Attenuation by Obstacles

Typical obstruction	Thickness (mm)	2,4 GHz signal attenuation (dB)	5 GHz signal attenuation (dB)	Notes
Pine wood with a hollow core	20	2	3	
Asbestos	8	3	4	
Foam board	8	3	4	
Ordinary wooden door	40	3	4	
Ordinary glass	40	10	15	
Thickened glass	8	4 - 5	7 - 9	Single layer
Solid wood door	12	5 -6	9 - 10	Double layer, no coating
Ordinary brick wall	12	8	10	Single layer
Roll-up door	-	8 - 10	15 - 20	One surface coated
Thickened brick wall	-	10 - 12	20 -25	Two surfaces coated
Concrete	120	10	20	
Load-bearing column	240	15	25	
Bulletproof glass	10	15	20	
Steel plate	240	25 - 30	30 - 35	Varies by moisture/rebar
Elevator	500	25 - 30	30 - 35	Varies by structure

NOTE: The above values are theoretical calculations. Deviations may occur in practical applications; actual test results are recommended to be used for verification.

History

Version	Date	Status
V1.1.1	September 2026	Publication